

Sun, Shuguo

Ph.D. Candidate

Phone: +1 (704)-600-7423

E-mail: sbs7733@psu.edu

Room 121, Institute of Energy and the Environment

Department of Mechanical Engineering

The Pennsylvania State University, University Park, PA 16802



Education

Ph.D. Mechanical Engineering, The Pennsylvania State University	08/2026-06/2027
Advisor: Prof. Jun Xu	(Expected)
Research topic: Thermal Hazards Science of Electrochemical Energy Storage Systems	
Ph.D. Student Mechanical Engineering, The University of Delaware	09/2023-07/2026
Advisor: Prof. Jun Xu	
M.S. Engineering Fluid Mechanics, Imperial College London	09/2019-09/2020
Advisor: Dr. Henry Burridge	
B.Eng. Built Environment and Energy Application Engineering, Chongqing University	09/2015-07/2019

Research Interests

- Durability and Sustainability of Lithium-Ion Battery Materials
- Electrochemical-thermal coupled modeling and computation
- Lithium-ion battery thermal safety and degradation

Working Experience

The Pennsylvania State University 09/2026-present

Research Assistant

- Degradation characterization and safety testing of lithium-ion battery materials
- Development of a battery thermal safety testing platform
- Development of physics-informed machine learning algorithms for predicting battery thermal performance

University of Delaware 09/2025-05/2026

Teaching Assistant, MEEG321: Materials Engineering and MEEG346: Thermal Laboratory

- Prepared course materials and laboratory instructions
- Facilitated class discussions and laboratory sessions
- Graded assignments, laboratory reports, and examinations

FM Global 05/2024-08/2024, 05/2025-07/2025

Intern

- LIB degradation and thermal runaway single-cell/module tests
- A numerical method to calibrate LIB thermal properties (experiments and modeling)
- Electrochemical-thermal coupling numerical modeling of LIB thermal runaway

Honors & Awards

Best Presentation Award, 249 th ECS Meeting	2026
K.M. Abraham Travel Award, 249 th ECS Meeting	2026
Graduate Student Travel Award, University of Delaware	2025
S. Subbiah & Sarah Robson Graduate Financial Support Award, University of Delaware	2025
14 th USNCM Student Travel Award, USNCM 2025	2025
Degree with Distinction, Imperial College London	2020
Outstanding Graduate, Chongqing University (TOP 5%)	2019
Merit Student, Chongqing University	2018
The third prize of Science & Technology Competition, Chongqing University	2017
The third prize of Comprehensive Scholarship, Chongqing University	2016
Gree Group Co., Ltd. Scholarship, Second Prize, Chongqing University	2016

Publications

Journal Publications (Published as first/co-first author: 6; Total citations: 186; H-index: 7)

- [1] **S. Sun**, B. Rui, S. Bahuguna, J. Zhou, F. Sayeed, J. Xu*, Electrochemo-mechanical-thermal dynamics of internal short circuits in batteries, *Energy Storage Materials*, 86 (2026) 105022. (IF=19.3)
- [2] **S. Sun**, W. Brown, X. Tan, B. Rui, G. Xiong, Y. Wang, D. Zeng*, J. Xu*, Aging Matters: How Degradation Pathways Govern Thermal Runaway in Lithium-Ion Batteries, *Journal of Energy Chemistry* 114 (2026) 10-21. (IF=15.6)
- [3] **S. Sun**, J. Xu*, Safety behaviors and degradation mechanisms of aged batteries: A review, *Energy Materials and Devices* (2024). (IF=9.9)
- [4] **S. Sun**, B. Rui, X. Tan, S. Bahuguna, J. Zhou, J. Xu*, Effects of electrolyte, state of charge, and strain rate on the mechanical properties of lithium-ion battery electrodes and separators, *Journal of Materials Chemistry A* 13 (2025) 20673-20687. (IF=9.5)
- [5] **S. Sun**, G. Xiong, Y. Wang, W. Brown, D. Zeng*, J. Xu*, Thermal runaway propagation behavior for aged lithium-ion batteries, *Fire Safety Journal* 162 (2026) 104795. (IF=3.8)
- [6] Y. Kawashima[†], **S. Sun**[†], D. Sumiya, A. Yonezu*, J. Xu*, Aging-Driven Mechanical Degradation of Lithium-Ion Battery Cathodes, *Journal of Power Sources* 663 (2026) 238949. (IF=8.4)
- [7] B. Rui, **S. Sun**, X. Tan, C. (Michael) Chak, L. Ma, J. Xu*, Mechanical abuse and safety in sodium-ion batteries, *Journal of Materials Chemistry A* 13 (2025) 12203–12215. (IF=9.5)
- [8] X. Tan, **S. Sun**, J. Xu*, Decoding the liquid-solid interplay for the mechanical behaviors of the battery materials, *Journal of Power Sources* 653 (2025) 237566. (IF=8.4)
- [9] K. Zhu, **S. Sun**, B. Rui, J. Xu*, Metal Particle Foreign Matter as a Hidden Driver of Battery Aging: A Defect-Informed Modeling Study, *ACS Applied Materials & Interfaces* 18 (2026) 23312-23326. (IF=7.8)
- [10] A. Bhowmick, B. Rui, **S. Sun**, X. Tan, B. Wei, Y. Hahn, S. Kulathu, V. Oancea, W. Lu, J. Xu*, Electrochemo-Mechanics Unlocks Hidden Dynamics of Lithium Plating under Stacking Pressure, *Nano Energy* 150 (2026) 111751. (IF=16.7)
- [11] X. Tan, N. Zeng, Y. Deng, **S. Sun**, B. Rui, J. Xu*, Early-cycle prediction of battery aging onset across chemistries, *Journal of Energy Chemistry* 105 (2025) 233–242. (IF=15.6)

* Corresponding author

[†] Equal contribution

- [12] X. Duan, D. Hu, W. Chen, J. Li, L. Wang, **S. Sun**, J. Xu*, Electrochemical Modeling of Fast Charging in Batteries, *Advanced Energy Materials* 14 (2024) 2400710. (IF=25.5)
- [13] B. Rui, S. Bahuguna, K. Zhu, J. Zhou, **S. Sun**, S. Chen, F. Sayeed, J. Xu*, Thermo-mechanical characterization of battery component materials with electrolyte coupling, *Journal of Power Sources*, 691 (2026) 240930. (IF=8.4)
- [14] M. Kajihara, J. Xiao, Y. Kawashima, **S. Sun**, X. Yuan, A. Yonezu, J. Xu*, State-of-health-dependent high-rate microparticle penetration of NMC532 and graphite electrodes, *Journal of Power Sources*, 693 (2026) 241219. (IF=8.4)

Oral Presentation in Conference

- [1] **S. Sun**, and J. Xu (2025). "Aging Path-Dependency in Thermal Hazards of Lithium-ion Batteries". Oral, The 14th USNCM Conference, Boston, MA, March 17.
- [2] **S. Sun**, and J. Xu (2025). "Aging Path-Dependency in Thermal Hazards of Lithium-ion Batteries". Oral, 2025 FM Fire Modeling Workshop, Norwood, MA, May 13.
- [3] **S. Sun**, and J. Xu (2025). "Electrochemo-Mechanical Coupling in Electrode Materials Under Electrolyte-Induced Interactions". Oral, IMECE 2025, Memphis, TN, November 17.
- [4] **S. Sun**, and J. Xu (2026). "Aging Path-Dependency in Thermal Hazards of Lithium-ion Batteries". Oral, WCX SAE World Congress Experience, Detroit, MI, April 15.
- [5] **S. Sun**, and J. Xu (2026). "Mechanism of Internal Short Circuits in Lithium-Ion Batteries under Mechanical Abuse". Oral, WCX SAE World Congress Experience, Detroit, MI, April 15.
- [6] **S. Sun**, and J. Xu (2026). "A Physics-Based Inverse Model for Thermal Hazards of Aged Lithium-Ion Cells". Oral, 2026 FM Fire Modeling Workshop, Norwood, MA, May 21.
- [7] **S. Sun**, and J. Xu (2026). "Physics-Informed Neural Networks for Predicting Thermal Runaway Trajectories across the Full Life Cycle". Oral, 249th ECS Meeting, Seattle, WA, May 27.
- [8] **S. Sun**, and J. Xu (2026). "Mechanism of Internal Short Circuits in Lithium-Ion Batteries Under Mechanical Abuse". Oral, 249th ECS Meeting, Seattle, WA, May 27.
- [9] **S. Sun**, and J. Xu (2026). "Multiphysics Mechanism of Internal Short Circuit in Batteries". Oral, ASME DRIVN 2026, Southfield, MI, September 16.

Patent

- [1] J. Yu, B. Zheng, P. Wang, Y. Li, and **S. Sun**, "A battery cooling plate, battery thermal management system, and vehicle". ZL202322070554.6 (Chinese Patent)

Projects

Lead researcher, Understanding the Thermal Hazards of the Aged Cell upon Abusive Scenarios, *FM Global* 09/2023-present

- Perform the battery degradation tests and thermal runaway tests
- Develop the electrochemical-thermal coupled model
- Develop the LIB thermal runaway testing platform

Lead researcher, High Voltage Battery Pack Deformation, *USCAR* 09/2025-03/2026

- Develop the detailed modeling framework of battery pack mechanical deformation

Lead researcher, Detailed Battery Cell Modeling and Component Characterization, *General Motors* 03/2024-05/2025

- Perform mechanical abusive loading tests and corresponding numerical modeling on pouch cells, including compression, indentation, and dynamic impact.
- Perform mechanical characterization on LIB components (*i.e.*, anode, cathode, and separator).

Lead researcher, Multiphysics Modeling and Characterization of the Mechanical Integrity of Lithium-ion Batteries: From cell, module to pack, *Altair* 09/2023-03/2024

- Develop the detailed modeling method of LIB mechanical abuse
- Develop the numerical modeling of mechanical tests of cell materials

Professional Affiliations

- Committee Member, *ASME* Technical Committee on *Electrochemical Energy Conversion and Storage (EECS)*.
- Session Chair for *IMECE 2025*, “*Electrochemical Energy Storage and Conversion Systems III*” and “*Electrochemical Energy Storage and Conversion Systems IV*”.
- Reviewer for *ASME conferences*, *SAE conferences*, *Energy Materials and Devices*, *J. Power Sources*, and *J. Electrochem. Energy Conv. Storage*.
- Student Member of *ASME*, *SAE*, *ECS*, and *IEEE*.